

Six-Axis Welding Robot (ROBOT-WELD) Equipment Manual and SOP

VINDHYA PRECISION WORKS

STANDARD OPERATING PROCEDURE & EQUIPMENT MANUAL

Asset Class: ROBOT-WELD (Six-Axis Arc Welding Robot)

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1.0 PURPOSE AND SCOPE This document defines the operational parameters, preventive maintenance schedules, fault diagnostics, consumable parts, and safety protocols for all six-axis articulated welding robots (Asset Class: ROBOT-WELD) installed in the Vindhya Precision Works production facility. Compliance with this procedure is mandatory for all assigned maintenance technicians and welding operators.

2.0 NORMAL OPERATING RANGES AND ALARM SETPOINTS Operation outside the parameters defined in Table 2.1 shall result in automated interlock tripping or operator intervention as specified.

Table 2.1: Operational Limits for ROBOT-WELD

Parameter	Normal Operating Range	Warning Alarm Setpoint	Critical Trip / Interlock Setpoint
Axis 1–3 Gearbox Temp	$+15^{\circ}\text{C}$ to $+52^{\circ}\text{C}$	$+60^{\circ}\text{C}$	$+70^{\circ}\text{C}$
Axis 4–6 Wrist Temp	$+15^{\circ}\text{C}$ to $+45^{\circ}\text{C}$	$+52^{\circ}\text{C}$	$+62^{\circ}\text{C}$
Welding Power Source Temp	$+20^{\circ}\text{C}$ to $+55^{\circ}\text{C}$	$+65^{\circ}\text{C}$	$+75^{\circ}\text{C}$
Arm Vibration (RMS)	0.2 mm/s to 1.8 mm/s	2.5 mm/s	3.5 mm/s
Shielding Gas Pressure	0.22 MPa to 0.28 MPa	$< 0.20\text{ MPa}$ or $> 0.30\text{ MPa}$	$< 0.15\text{ MPa}$ or $> 0.35\text{ MPa}$
Pneumatic Brake Supply	0.55 MPa to 0.65 MPa	$< 0.50\text{ MPa}$	$< 0.42\text{ MPa}$
Welding Current (Arc On)	80 A to 320 A	$< 70\text{ A}$ or $> 335\text{ A}$	$< 50\text{ A}$ or $> 350\text{ A}$
Servo Bus Voltage	560 VDC to 640 VDC	$< 540\text{ VDC}$ or $> 660\text{ VDC}$	$< 500\text{ VDC}$ or $> 680\text{ VDC}$

3.0 PREVENTIVE MAINTENANCE SCHEDULE BY RUNNING HOURS

Maintenance tasks are cumulative. When a higher hour milestone is reached, all lower-hour tasks shall also be executed. Running hours are tracked via the robot controller internal PLC register `R[902]`.

3.1 Interval: Every 250 Operating Hours (Shift Level) 1. Inspect the welding torch contact tip and gas nozzle for spatter accumulation; clean using pneumatic reamer station and apply anti-spatter fluid (VPW part W-SPAT-01). 2. Check the wire feed drive rolls for particulate debris and groove wear; blow out with dry instrument air at $\leq 0.3 \text{ MPa}$. 3. Verify operation of the TCP (Tool Center Point) calibration routine using the fixed reference jig. Deviation exceeding $\pm 0.5 \text{ mm}$ requires reprogramming. 4. Inspect external cabling, teach pendant umbilical, and torch dress package for chafing or insulation cuts.

3.2 Interval: Every 1,000 Operating Hours (Weekly Level) 1. Grease the wire feeder push-pull gear assembly using lithium complex grease (VPW part LUB-WELD-08). 2. Inspect mechanical hard stops on Axes 1 through 6 for impact scoring or deformation. 3. Clean the cabinet air intake filters on the controller and welding power source using vacuum extraction. 4. Check battery backup voltage for absolute encoders; replace if voltage reads below 2.85 VDC (Nominal 3.0 VDC).

3.3 Interval: Every 4,000 Operating Hours (Semi-Annual Level) 1. Inspect and purge harmonic drives on Axes 4, 5, and 6; replenish with synthetic harmonic grease (VPW part LUB-ARM-12) per volume matrix in Section 6.0. 2. Verify servo motor brake holding torque on all six axes using the controller static brake test utility. Minimum acceptable holding torque is 125% of rated motor stall torque. 3. Inspect internal hollow-wrist cable harness for twisting or insulation fatigue. 4. Calibrate analog shielding gas mass flow controller against secondary standard meter.

3.4 Interval: Every 12,000 Operating Hours (Annual Overhaul Level) 1. Complete flush and refill of RV reduction gearboxes on Axes 1, 2, and 3 using synthetic gear oil (VPW part LUB-RV-20). 2. Replace timing belts on Axis 2 and Axis 3 drives. Tensioning must be set to 45 Hz vibration frequency using acoustic tension meter. 3. Perform full insulation resistance (Megger) test on all servo motor windings at 500 VDC ; minimum acceptable resistance is

4. Replace main interface relays and safety contactors inside the controller cabinet.

4.0 FAULT-CODE TABLE AND DIAGNOSTICS The following table details named fault conditions generated by the ROBOT-WELD controller, their root causes, and mandatory corrective actions.

Table 4.1: Fault-Code Diagnostics

Fault Code & Name	Probable Cause	Corrective Action
ERR-WLD-101: Wire Burnback Detected	Contact tip fusion due to erratic wire feed speed, incorrect burnback timer setting, or low shielding gas flow causing immediate arc instability.	Inspect contact tip bore diameter; replace if worn by > 15%. Verify gas flow rate at nozzle using rotameter. Clean or replace wire feed drive rolls. Adjust burnback time parameter <code>P[14]</code> by +0.05 s increments.
ERR-WLD-102: Servo Overcurrent Axis 3	Mechanical binding in Axis 3 harmonic drive, excessive payload inertia, or debris lodged in the lower arm casting cavity.	Perform manual brake release and hand-rotate Axis 3 to check for binding. Inspect gearbox backlash and housing temperature. Clear any foreign obstruction. If binding persists, replace harmonic reducer assembly.
ERR-WLD-103: Gas Pressure Interlock Open	Primary regulator failure, upstream supply bottle depletion, or pinched flexible feed line inside the upper arm dress pack.	Check primary regulator delivery pressure. Verify supply manifold pressure. Inspect flexible polyurethane gas line for kinks or crushing along the J3-J6 arm routing. Replace damaged hose sections.
ERR-WLD-104: Encoder Communication Loss J5	Pulse coder cable failure, loose MIL-spec connector at J5 motor base, or electrical noise coupling from welding high-frequency arc starter.	Power down controller. Inspect, clean, and reseal the 20-pin encoder connector at Axis 5 motor. Check shield grounding wire integrity at both ends. Verify continuity of internal wiring harness cores.
ERR-WLD-105: Collision Sensor Tripped	Mechanical torch collision with fixture, workpiece distortion, or incorrect approach vector causing pneumatic switch separation.	Inspect torch alignment and TCP calibration. Clear physical obstruction. Reset the pneumatic breakaway switch plunger. If optical proximity sensor on collision plate is misaligned, readjust bracket and retighten fasteners.
ERR-WLD-106: Arc Failure to Ignite	Uncleaned base metal surface, excessive distance between contact tip and workpiece, or worn contact tip producing unstable arc initiation.	Inspect workpiece joint for rust, oil, or thick mill scale; re-clean as per welding specification. Verify TCP height parameter. Replace pitted or oversized contact tip. Check earth ground clamp connection to welding table.

5.0 TORQUE SPECIFICATIONS All threaded fasteners in structural, motor, and transmission locations on ROBOT-WELD units shall be tightened using calibrated torque wrenches in accordance with Table 5.1. Threads must be clean and dry unless threadlocker application is specified.

Table 5.1: Fastener Torque Matrix

Fastener Location / Description	Thread Size & Grade	Torque Limit (\$\text{Nm}\$)	Application Notes
Axis 1 Base Mounting Bolts	M16, Grade 12.9	\$210\text{ Nm} \pm 5\%\$	Apply Loctite 271; star pattern tightening sequence.
Axis 2 & 3 Reducer Flange Bolts	M12, Grade 12.9	\$95\text{ Nm} \pm 3\%\$	Apply Loctite 243; cross-torque in 3 stages.
Axis 4, 5, 6 Wrist Housing Bolts	M6, Grade 10.9	\$12\text{ Nm} \pm 2\%\$	Apply Loctite 243; titanium-coated housing threads.
Servo Motor Mounting Fasteners	M8, Grade 10.9	\$28\text{ Nm} \pm 3\%\$	Dry assembly.
Welding Torch Clamp Bolt	M8, Grade 8.8	\$22\text{ Nm} \pm 5\%\$	Ensure insulated collar is seated prior to torque.
TCP Reference Jig Retaining Screws	M5, Grade 8.8	\$8\text{ Nm} \pm 5\%\$	Dry assembly.

6.0 CONSUMABLES AND SPARE PARTS LIST Technicians shall only utilize genuine or engineering-approved replacement parts listed in Table 6.1. Substitution of unapproved parts voids equipment certification.

Table 6.1: Approved Consumables and Spares

Item Description	VPW Part Number	Replacement Frequency / Threshold
Contact Tip (1.2 mm CuCrZr)	W-TIP-12-01	Every 50 operating hours or upon arc instability
Gas Nozzle (Conical, 16 mm bore)	W-NOZ-16-03	Every 200 operating hours or heavy spatter bridging
Torch Swan Neck Assembly (30° offset)	W-NECK-30-02	Every 4,000 operating hours or physical distortion
Wire Feed Drive Roll (1.2 mm V-Groove)	W-ROL-12-V	Every 2,000 operating hours
Harmonic Grease (Arm Axes 4–6)	LUB-ARM-12	Every 4,000 operating hours (45 ml per cavity)
RV Gearbox Oil (Base Axes 1–3)	LUB-RV-20	Every 12,000 operating hours (850 ml per reducer)
Encoder Backup Battery Pack (3.0 VDC)	ELE-BAT-99	Every 8,000 operating hours or 2 years
Internal Dress Pack Harness Assembly	W-HARN-ROB-06	Every 16,000 operating hours or outer sheath rupture
Pneumatic Collision Switch Module	W-COL-SW-04	As required upon failure of ERR-WLD-105

7.0 SAFETY-CRITICAL CORRECTIVE ACTIONS (LOCKOUT & PERMIT REQUIRED) Execution of the following maintenance and corrective actions alters the safety envelope of the ROBOT-WELD asset. These tasks strictly require the issuance of a formal Maintenance Work Permit (Form VPW-MWP-09), complete execution of Lockout/Tagout (LOTO) per Plant Procedure VPW-SAF-001, and verification of zero energy state prior to tool contact.

1. Replacement or internal wiring repair of Axis 1 through Axis 3 servo motors or electromagnetic brakes. (Involves gravitational hazard; mechanical drop pin must be inserted into the Axis 2 transport locking bore before work commences).
2. Disassembly, repair, or replacement of Axis 1, 2, or 3 reduction gearboxes or timing belt drive assemblies.
3. Welding power source internal high-voltage capacitor bank discharge and internal inverter module replacement. (Requires verification of $< 15\text{ VDC}$ residual bus voltage using a CAT III / CAT IV calibrated multimeter).
4. Replacement of the internal hollow-wrist cable harness (Dress Pack), involving the disconnection of main power, control, gas, and water lines through the J3 to J6 arm assembly.
5. Modification, bypassing, or hardwire testing of the safety fence interlock circuit, optical light curtains, or emergency stop chain connected to the robot controller safety PLC.

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